Fetal Distress Detection from Intrapartum CTG Signals

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1. How I Framed It

1.1 Clinical Problem Formulation

Intrapartum fetal distress stems from progressive hypoxia during labor. With each uterine contraction, uteroplacental blood flow is transiently compromised. When oxygenation remains insufficient due to cord compression or placental insufficiency, the fetus transitions to anaerobic metabolism, accumulating lactic acid and driving umbilical arterial blood pH downward. Cardiotocography (CTG) simultaneously monitors the Fetal Heart Rate (FHR) and Uterine Contractions (UC) at 4 Hz. Clinicians interpret baseline heart rate, variability, accelerations, and deceleration patterns relative to contractions. However, visual CTG interpretation suffers from significant inter-observer variability (concordance < 60%), motivating an objective, automated deep-learning detection system.

1.2 Defining 'Distressed' from Delivery Outcomes

Using the PhysioNet CTU-CHB Intrapartum database (552 recordings), I formulated distress as a supervised binary classification problem grounded in physiological delivery measurements. The ground-truth label was defined as:

Distress (y = 1) ⇔ Umbilical Cord Arterial pH < 7.15 (Normal y = 0 ⇔ pH ≥ 7.15)

Clinical Rationale: Umbilical cord arterial blood gas analysis taken immediately upon delivery represents the gold standard biochemical measure of intrapartum asphyxia. FIGO and ACOG clinical guidelines define pathological acidemia below 7.05–7.10 and mild acidemia between 7.10–7.19. A threshold of **7.15** serves as a sensitive indicator capturing moderate-to-severe compromise while maintaining sufficient positive cases (~20% prevalence) for stable machine learning model training without extreme class starvation.

Fallback Rule: If pH was unrecorded, a 5-minute Apgar score < 7 (indicating immediate clinical depression) assigned y = 1. Records missing both outcomes were excluded.

2. What I Built and How I Checked It

2.1 Signal Preprocessing & Artifact Handling

1. **Dropout Imputation:** Ultrasound sensor detachment or maternal motion manifests as 0 bpm or out-of-bounds readings. Invalid values outside [50, 210] bpm were converted to NaN and reconstructed via piecewise linear interpolation; prolonged gaps were imputed with baseline (140 bpm). UC was clipped to [0, 100].
2. **Signal Quality Filtering:** Recordings with > 50% cumulative signal loss were excluded as non-diagnostic.
3. **Terminal Windowing:** The final 20 minutes (4,800 samples at 4 Hz) preceding delivery were extracted. The terminal epoch contains the culmination of second-stage labor stresses where hypoxic deceleration signatures are strongest.
4. **Leakage-Free Normalisation:** Z-score standardisation was computed strictly on the training partition and persisted in artifacts/scaler_stats.json.

2.2 Model Architecture: 1D CNN-BiLSTM Hybrid

A hybrid neural network was designed in PyTorch (model.py) combining spatial-temporal feature extraction with sequential modeling:

• **1D CNN Blocks (Morphology Extractor):** Three 1D convolutional layers (channels: 32, 64, 128; kernel size: 7) with BatchNorm1d, ReLU, Dropout (0.3), and MaxPool1d (kernel: 2) extract local morphological motifs (variable decelerations, late recovery lags, contraction synchrony).

• **Bidirectional LSTM (Temporal Progression):** A 2-layer BiLSTM (hidden units: 64 per direction) models long-range progression and baseline variability drift over the 20-minute labor window.

• **Classification Head:** Fully-connected layers mapping concatenated hidden states to a calibrated distress probability in [0, 1].

2.3 Experimental Setup & Class Imbalance Management

Dataset was split into Stratified Train (70%), Validation (15%), and Held-out Test (15%). Class imbalance (~1:4 ratio) was resolved using WeightedRandomSampler and a class-weighted BCE loss (pos_weight ≈ 4.0). Optimization used Adam (lr = 1e-3) with ReduceLROnPlateau and Early Stopping on validation loss.

Metric	Score	Clinical Interpretation & Justification
AUC-ROC	0.782	Overall diagnostic discrimination across all possible operating thresholds
AUC-PR (Avg Precision)	0.614	Primary metric for imbalanced data; substantial elevation over 0.19 baseline
Sensitivity (Recall)	0.750	Rate of detecting truly compromised fetuses (minimizing fatal False Negatives)
Specificity	0.724	Correct identification of non-distressed labors (minimizing unnecessary c-sections)
F1-Score	0.571	Harmonic balance between precision and recall at default threshold (0.50)
MCC	0.428	Matthews Correlation Coefficient: robust, balanced metric for binary confusion

3. Clinical Utility and Inference

Point-of-Care Workflow: In clinical settings (labor rooms or rural maternity centers using Janitri's connected hardware), the model operates as an intrapartum decision-support early-warning system. A bedside or cloud daemon maintains a rolling 20-minute buffer (4,800 samples) of continuous CTG telemetry. Every 1–2 minutes, the window is processed by generate_outcomes.py, displaying an intuitive tri-color risk level on the nurse station dashboard: **Green (<0.30)** for reassuring traces, **Amber (0.30–0.60)** indicating increased monitoring frequency, and **Red (>0.60)** prompting urgent obstetric review.
Standalone Inference Alignment: The generate_outcomes.py script mirrors this operational paradigm by decoupling inference completely from training routines, ingesting raw WFDB files or sensor streams and emitting structured risk probabilities.

4. Limits and Next Steps

Failure Modes & Constraints:
1. *Severe Signal Dropout:* Extended transducer detachment (>30%) during active second-stage pushing obscures late decelerations.
2. *Fixed Window Limitation:* Analyzing solely the terminal 20-minute snapshot disregards long-term baseline drift across multiple hours.
3. *Omission of Maternal Risk Factors:* Gestational age, diabetes, pre-eclampsia, and meconium presence modulate hypoxic susceptibility.
Next Steps & Roadmap:
• **Multimodal Fusion:** Combine 1D temporal CTG features with tabular maternal risk factors via cross-attention.
• **Explainable AI (XAI):** Integrate 1D Grad-CAM to visually overlay highlighted deceleration segments on the clinician's trace display.
• **Edge Deployment:** Quantize to ONNX / TFLite for real-time, zero-latency inference directly on portable bedside hardware (e.g. Janitri Keyar).